

Diego Arnaldo Azuela Rosas

Software Engineer - Embedded and Systems

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PROFESSIONAL SUMMARY

Software engineer with 3 plus years at Tesla building production Go and Python systems across high-voltage energy storage, humanoid robotics, and vehicle platforms. Primary author and maintainer of the shared Go instrument-driver and hardware-abstraction libraries (375 plus commits across two core repos, 9 instrument vendors) that test systems in Fremont, Lathrop, Shanghai, and contract-manufacturer sites depend on. Depth in UDS / ISO 14229 diagnostics and firmware download, CAN, gRPC services, concurrent state-machine architectures with strict timeout and fault semantics, and the SQL and analytics layer that turns test data into yield decisions. Published researcher at Harvard Medical School.

PROFESSIONAL EXPERIENCE

Launch Engineer Tesla, Inc. Mar 2023 - Present California

Energy Division - Software Development

- Instrument Driver and HAL Platform (Golang):** Primary author and top contributor of two shared Go libraries, a VISA/SCPI instrument-driver layer and a hardware-abstraction layer, with 375 plus commits, consumed by test systems across multiple factories. Built and maintained drivers for 9 instrument vendors (Chroma, Keysight, Keithley, TDK, Itech, Magna, Siglent, NI, WAGO). Designed the 3-layer architecture (serial and VISA transport, typed SCPI command library, test orchestration) and approximately 200 SCPI commands for the Chroma hipot family, and lifted instrument config into the shared HAL so new stations integrate by config, not code.
- Megapack Thermal System (Golang, SCPI, ECU, Linux):** Top software contributor on an HV test system projected to save USD 104M per year. Delivered a multi-system tester covering Hipot, LV, HV, and pneumatic actuation for coolant flow and leak detection, and owned thermal and coolant-flow health monitoring. Co-authored the Pilz safety program and obtained safety buyoffs at 220V and 250V, and implemented graceful exit routines for edge-case HV discharge and power-loss events.
- Safety Controller FVT (Golang, UDS, gRPC, CAN):** Co-maintainer of the functional verification tester for a safety-controller PCBA: a state-machine application (init, comms, flash, alarms, latch drive) fronting a gRPC and protobuf bridge to CAN and UDS services. Implemented the full ISO 14229 flashing sequence, covering security access, bootloader entry by ECU reset with tester-present, a 21-minute application download under a dedicated 30-minute context, and genealogy writes (part number, serial, assembly and usage IDs) by DID. Hardened measurement: median-of-N sampling with missing-sentinel rejection and CAN-MUX-aligned sampling to prevent aliasing.
- BMS Communication Station (Python, CAN):** CAN bus polling station catching broken BMS chain-communication faults invisible to existing EOL testers, preventing approximately USD 2M per week in throughput loss on a 4,800-unit per week line.
- Bi-Directional Inverter Station:** 24V signal injection and CAN-over-Ethernet readback validating the ECU communication chain; improved EOL first-pass yield 5 percent, saving approximately USD 55K per week.
- Manufacturing Data and OEE Analytics (SQL, Power BI, Python):** Architected a PLC-to-SQL diagnostic pipeline across 18 HV converter testers, achieving 99.6 percent first-pass yield, highest in Tesla Energy division. Maintain a 100 plus query SQL analytics library covering failure-mode analysis, SCADA availability, and OEE.
- Telemetry Daemon (Linux, OPC UA):** Linux daemon monitoring 200 plus Siemens OPC UA nodes with automated fault alerting and recovery orchestration, reducing downtime 65 percent.
- ECU Validation Station (Python):** Firmware validation station for fault isolation and compliance at 200K plus units per year scale. Recovered USD 2.4M per week in throughput and cut Japan compliance tester cost USD 450K.
- Test Platform Tooling and Consolidation (Golang, Python):** Consolidated a duplicated test runner into one shared repository used across all sites. Built internal tooling used across test engineering: a Go parser generating safety documentation from Pilz PNOZ files, DBC revision diffing, and automated Teams alerting. Added barcode-scanner error handling preventing untested boards from passing.
- Golang Systems and NPI:** Re-integrated Go test systems for Shanghai supply lines, improving uptime from 75 to 90 percent. Led Megapack 3 NPI, enabling a 215-unit (900 MWh) single-week delivery.

Humanoid Robotics - Optimus

- Optimus Charger EOL (Python, CAN, PCAN, AWS S3):** Drove Charger first-pass yield from 0 percent to 80 percent during NPI by debugging ECU firmware via CANape, DBC decoding, and HEX analysis. Integrated AWS S3 as a centralized trace store for CAN logs, Cut station cycle time via init-time upload setup, trace compression, and structured JSON logging for machine-parseable failures.

Vehicles Division

- Vehicles Division:** Led BMS integration and wireless charging test deployment at contract manufacturer sites in Mexico and the USA. Resolved validation gaps and reduced program cost.

RESEARCH EXPERIENCE

Research Intern Harvard Medical School, Brigham and Women's Hospital Jul 2021 - Feb 2022 Cambridge, MA

- Developed Python control software for a 6-DOF robotic bioprinter using OpenCV depth sensing and computational geometry path planning. Validated deposition accuracy end-to-end. Advanced Materials Technologies, 2024.
- Built an electrochemical biosensor data acquisition and signal processing pipeline integrated into a microfluidic bioreactor for real-time stem cell monitoring. ACS Sensors, 2024.

PUBLICATIONS

Jeong et al. Intelligent In Situ Printing of Multimaterial Biopinks for First-Aid Wound Care Guided by Eye-In-Hand Robot Technology.

Rezaei et al. Noninvasive and Continuous Monitoring of On-Chip Stem Cell Osteogenesis Using a Reusable Electrochemical Immunobiosensor.

EDUCATION

Tecnologico de Monterrey (ITESM) B.Sc. Mechatronics Engineering, GPA 3.73, 2017-2023

Penn State University B.Sc. Mechanical Engineering, GPA 3.75, Dean's List, 2016-2017

TECHNICAL SKILLS

Languages: Golang, Python, SQL, C++, Shell, Bash, LabVIEW, TestStand, MATLAB

Software Engineering: Go concurrency (goroutines, context, channels), gRPC and protobuf, REST API integration, state-machine architecture, driver and library design, code review, Git branching, semantic versioning and release tagging

Protocols: CAN bus, UDS / ISO 14229 (DIDs, security access, bootloader, FWDL), CANape, DBC, PCAN, HEX, OPC UA, SCPI and VISA, Modbus TCP, I2C, SPI, PLC, ECU firmware

Infrastructure: Linux (systemd), AWS S3, MySQL, Docker, CI/CD, pytest, Git, Ansible, Power BI, OpenCV

Methods: SPC, Gage R and R, fault injection, FMEA, NPI, DVT, PVT, V and V, root cause analysis

Languages (spoken): English (TOEFL 640), Spanish (native), French and Italian (intermediate)